ECA08 – Analog and Digital Communication

CO1 Design Task Assignment: AM-Based Communication Link Design, from Modulation Selection to Receiver Deployment

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Selection → Design → Realization → Verification → Deployment decision. All power values are for single-tone modulation with m = 0.7 unless stated otherwise, and powers are taken across a 1 Ω normalized load unless a 50 Ω load is mentioned.

Stage 1: Interpreting the Communication Design Requirement

1.1 Basic communication system

Every communication system has the same basic blocks. For this problem they map as follows.

Studio voice	-->	Input transducer (microphone)	-->	Transmitter (modulator, PA, antenna)	-->	CHANNEL (air + noise)	-->	Receiver (RF/IF, detector)	-->	Output transducer (loudspeaker)	-->	Listener
(source)												

- Source and input transducer: the studio voice is converted to an electrical signal by a microphone. Voice has useful energy up to about 4 kHz, so f_m(max) = 4 kHz.
- Transmitter: conditions the signal (amplify, low pass filter), generates the carrier, modulates it and amplifies it to the required power.
- Channel: free space between the antennas. It attenuates the signal and adds AWGN, atmospheric noise and interference from other stations.
- Receiver: picks the wanted station out of many, converts it to a fixed frequency, demodulates it and amplifies the audio.
- Output transducer: loudspeaker or headphone.

1.2 Why modulation is needed

Sending the 0–4 kHz voice directly through the air (baseband transmission) is not practical, for three reasons.

1. Antenna size. An efficient antenna needs a length of roughly λ/4. At 4 kHz, λ = c/f = 3×10^8 / 4000 = 75 km, so λ/4 is about 18.75 km. At a 1 MHz carrier, λ = 300 m and λ/4 = 75 m, which is large but achievable, and is why the MF band uses tall vertical masts.
2. No multiplexing. Every voice occupies the same 0–4 kHz band, so two stations would overlap completely and the receiver could not separate them. Modulation moves each station to its own carrier frequency.
3. Noise and range. Low frequency signals radiate poorly and are easily disturbed by interference. A carrier in the MF band propagates as a ground wave and is easier to amplify and filter.

1.3 Message bandwidth and transmission bandwidth

Message bandwidth W = 4 kHz.

Scheme	Theoretical transmission bandwidth	Value for W = 4 kHz
Conventional AM	2W	8 kHz
DSB-SC	2W	8 kHz
SSB-SC	W	4 kHz
VSB-SC	W + f_v (f_v = vestige width)	about 4.5 kHz if f_v = 0.5 kHz

1.4 Channel planning idea

Because each station is moved to a different carrier, the RF band can be shared by frequency division multiplexing (FDM). Stations use carriers separated by a fixed spacing, so their spectra sit side by side without overlapping. The receiver picks one using a tunable filter.

1.5 Number of 20 kHz channels in 540–1600 kHz

Available band = 1600 − 540 = 1060 kHz.

Number of channels = 1060 / 20 = 53 (idealised).

If the 1 MHz candidate carrier must be one of the carriers, the carriers fall on the grid 1000 ± 20n kHz. The first carrier that keeps its lower sideband inside the band is 560 kHz (sidebands 556–564 kHz) and

the last is 1580 kHz (1576–1584 kHz). That gives 22 channels below 1 MHz, the 1 MHz channel itself, and 29 above, so 52 channels. One channel is lost to keep 1 MHz on the grid, which I accept.

1.6 Power calculation for $P_c = 10$ W, $m = 0.7$

Total AM power:

$$P_t = P_c\left(1+\frac{m^2}{2}\right) = 10\left(1+\frac{0.49}{2}\right) = 12.45\ \text{W}$$

Total sideband power = $P_c m^2/2 = 10 \times 0.245 = 2.45$ W, so each sideband carries 1.225 W.

Efficiency $\eta = 2.45 / 12.45 = 0.1968$, so only 19.68 % of the transmitted power carries information. The other 80.3 % (10 W) is the carrier.

Peak envelope power (PEP) = $P_c (1+m)^2 = 10 \times 1.7^2 = 28.9$ W. The power amplifier must stay linear up to this peak, not only at 12.45 W average.

For a 50 Ω antenna feed: $A_c = \sqrt{(2 \times 10 \times 50)} = 31.6$ V peak, and each sideband has amplitude $m A_c / 2 = 11.07$ V.

1.7 Receiver requirements

Parameter	Meaning	Requirement for this link
Sensitivity	Smallest input signal giving the required output SNR	Output SNR of 20 dB for clean voice
Selectivity	Ability to reject neighbouring channels	Pass about 8–9 kHz, strongly reject stations 20 kHz away
Fidelity	Faithful reproduction of the audio	Flat response up to 4 kHz, low distortion
Stability	Local oscillator and IF should not drift	Drift small compared with 1 kHz so tuning stays on station
Image rejection	Rejecting the second frequency that also mixes to the IF	More than about 35 dB over the whole band
IF consideration	Fixed frequency where most gain and filtering is done	Outside the RF band, and low enough for a narrow IF filter

1.8 Practical constraints (low-power educational prototype)

- The 10 W carrier is the design specification. On the bench the prototype runs at milliwatt level with the same m = 0.7. Power ratios do not change with scale.
- Transmitting on 540–1600 kHz needs a licence, so the lab link is wired through a dummy load or a short coupling loop.
- Components should be common: op-amps, 455 kHz ceramic filters, ferrite coils, a variable capacitor and a balanced modulator IC such as the MC1496.
- Phase shift networks and sharp SSB filters are hard to build in a college lab. This pushes the choice towards simple generation and detection.
- A low-cost receiver needs a simple detector. A receiver that needs a synchronised local carrier costs more and is harder to tune.

Stage 1 deliverable: Communication Design Requirement Statement

The link shall carry voice limited to 4 kHz over the 540–1600 kHz band with carriers on a 20 kHz grid anchored at 1 MHz (52 usable channels). Each station shall occupy no more than 8 kHz, leaving at least 12 kHz of guard band between neighbours. The transmitter shall deliver a 10 W unmodulated carrier at m = 0.7 (total 12.45 W, PEP 28.9 W). The receiver shall give at least 20 dB output SNR, reject adjacent channels and image channels by more than 35 dB, and keep audio response flat up to 4 kHz. The design shall use components available in an undergraduate lab.

Stage 2: Investigating and Selecting the Modulation Scheme

2.1 Time-domain and frequency-domain expressions

Let m(t) be the message with bandwidth W = 4 kHz, M(f) its spectrum, and c(t) = A_c cos(2π f_c t) the carrier.

$$\text{AM:}\quad s_{AM}(t)=A_c\,[1+k_a m(t)]\cos(2\pi f_c t),\qquad \mu = k_a\,|m(t)|_{max}$$

$$S_{AM}(f)=\tfrac{A_c}{2}[\delta(f-f_c)+\delta(f+f_c)]+\tfrac{A_c k_a}{2}[M(f-f_c)+M(f+f_c)]$$

$$\text{DSB-SC:}\quad s_{DSB}(t)=A_c\,m(t)\cos(2\pi f_c t),\qquad S_{DSB}(f)=\tfrac{A_c}{2}[M(f-f_c)+M(f+f_c)]$$

$$\text{SSB-SC (USB):}\quad s_{SSB}(t)=\tfrac{A_c}{2}\left[m(t)\cos(2\pi f_c t)-\hat m(t)\sin(2\pi f_c t)\right]$$

$$\text{VSB-SC:}\quad s_{VSB}(t)=\tfrac{A_c}{2}\left[m(t)\cos(2\pi f_c t)\mp m_Q(t)\sin(2\pi f_c t)\right],\quad S_{VSB}(f)=\tfrac{A_c}{2}[M(f-f_c)+M(f+f_c)]H(f)$$

Here $\hat{m}(t)$ is the Hilbert transform of m(t) (every frequency shifted by −90°), and $m_Q(t)$ is m(t) passed through the filter that gives the vestige. The VSB filter must satisfy $H(f-f_c) + H(f+f_c) = \text{constant}$ for $|f| \leq W$, so that the demodulated message has no amplitude distortion. This means the response must roll off with odd symmetry about f_c, which is why the vestige exactly makes up for the part of the sideband that was cut off.

For tone modulation $m(t) = A_m \cos(2\pi f_m t)$ with $\mu = k_a A_m$:

$$s_{AM}(t)=A_c\cos(2\pi f_c t)+\tfrac{\mu A_c}{2}\cos 2\pi(f_c+f_m)t+\tfrac{\mu A_c}{2}\cos 2\pi(f_c-f_m)t$$

2.2 Power calculations ($P_c = 10$ W, $\mu = 0.7$)

Quantity	Value	Working
Modulation factor	0.7	given
Carrier power	10 W	given
Power in each sideband	1.225 W	$P_c \mu^2/4 = 10 \times 0.49/4$
Total sideband power	2.45 W	$P_c \mu^2/2$
Total AM power	12.45 W	10 + 2.45
Efficiency	19.68 %	2.45/12.45
Efficiency at $\mu = 1$ (best case)	33.3 %	$\mu^2/(2+\mu^2) = 1/3$

2.3 Power advantage of DSB-SC and SSB-SC over AM

The information is only in the sidebands. If the carrier is suppressed, the same sideband power (2.45 W) can carry the same information.

Scheme	Power needed for the same message	Saving relative to AM (12.45 W)	Power ratio AM : scheme
AM	12.45 W	–	1

Scheme	Power needed for the same message	Saving relative to AM (12.45 W)	Power ratio AM : scheme
DSB-SC	2.45 W	10.0 W (80.3 %)	5.08 (7.06 dB)
SSB-SC	1.225 W	11.225 W (90.2 %)	10.16 (10.07 dB)
VSB-SC	between 1.225 W and 2.45 W, close to SSB for a small vestige	about 85–90 %	about 5 to 10

Peak envelope power for the same single tone is also very different: AM 28.9 W, DSB-SC about 4.9 W, SSB-SC about 1.225 W. So the power amplifier of an AM transmitter is bigger and more expensive, not only the average power.

2.4 Comparison of the four schemes

Feature	AM	DSB-SC	SSB-SC	VSB-SC
Bandwidth (W = 4 kHz)	8 kHz	8 kHz	4 kHz	about 4.5 kHz
Carrier	Transmitted	Suppressed	Suppressed	Suppressed or reduced
Power efficiency	Low (19.7 %)	High	Highest	High
Generation	Simple (any nonlinear or product stage)	Balanced modulator	Filter method or phase-shift method; sharp filter or wideband 90° network	DSB followed by an asymmetrical filter
Detection	Envelope detector (diode + RC)	Coherent (needs synchronous carrier)	Coherent	Coherent, or envelope if a large carrier is added
Synchronisation	Not needed	Phase and frequency must match	Frequency must match closely	Needed

Feature	AM	DSB-SC	SSB-SC	VSB-SC
Effect of phase error φ	None	Output ∝ cos φ, goes to zero at 90°	Phase distortion in audio	Quadrature distortion
Receiver cost	Very low	High	High	Moderate to high
Typical use	MF broadcast voice	Stereo subcarriers, some data links	Narrowband voice, amateur radio	Analog TV video

2.5 Why VSB is used for analog television

A video signal has a bandwidth of about 4.2 MHz (NTSC) to 5 MHz (PAL-B/G, used in India), and contains important low frequency content (down to DC, for brightness).

- DSB would need twice the bandwidth (about 8.4–10 MHz per channel), which wastes spectrum.
- SSB would save the bandwidth, but its sharp filter cannot cut a signal with strong components near DC. An ideal filter that removes one sideband right at the carrier is not realisable and would distort the low frequencies and the picture phase.
- VSB sends one full sideband and only a vestige (a small part) of the other. The filter has a gradual roll-off across the carrier, which is practical to build. NTSC uses a vestige of about 0.75 MHz and fits video plus sound in 6 MHz; PAL-B/G fits it in 7 MHz.

The role of the vestige is to replace the part of the missing sideband near the carrier so the low frequencies are still reproduced correctly after detection. The response around f_c is shaped so that H(f−f_c) + H(f+f_c) = constant. In television the transmitted carrier is large, so the TV receiver uses a simple envelope-type (quasi-synchronous) detector at low cost, again an AM-style trade-off made for millions of receivers.

Design Decision 1

I recommend conventional AM (double sideband with full carrier, m = 0.7) with envelope detection, because it is a broadcast link where one transmitter serves many receivers, so the cost and complexity should sit at the transmitter and not in every receiver.

Supporting numbers:

1. Bandwidth fits. AM needs 8 kHz, and the given spacing is 20 kHz, so 12 kHz of guard band remains. SSB would reduce the occupied bandwidth to 4 kHz, but with a fixed 20 kHz spacing that saving cannot be used to add channels, so SSB's main advantage is not needed here.

2. Power cost is accepted at one place only. AM costs 12.45 W against 2.45 W (DSB-SC) or 1.225 W (SSB-SC) for the same information, an extra 10 W at the transmitter. For a single community transmitter this is acceptable. For many receivers, a coherent detector in each would cost far more than 10 W of extra transmitted power.
3. Synchronisation is avoided. Coherent detection of DSB-SC gives output proportional to cos φ: a 60° phase error halves the output (−6 dB) and a 90° error gives zero. For SSB-SC, a frequency error of even 100 Hz shifts all audio by 100 Hz, which makes speech sound unnatural. Each receiver would need an oscillator stable to about 50 ppm or better (50 Hz at 1 MHz) or a PLL/Costas loop. The envelope detector has none of these requirements.
4. The specification itself assumes it. The assignment fixes $\mu = 0.7$ and a 10 W carrier, which only makes sense for a transmitted-carrier scheme.
5. Constructability. A diode detector plus a ceramic IF filter is possible in a lab. A phase-shift SSB generator with accurate 90° shift over 300 Hz–4 kHz is hard to realise.

I do not claim AM is the best in power efficiency; SSB-SC is clearly better at 10 dB. For this problem the receiver complexity and the 20 kHz spacing outweigh that advantage.

Stage 3: RF Channel Planning and Transmitter Design

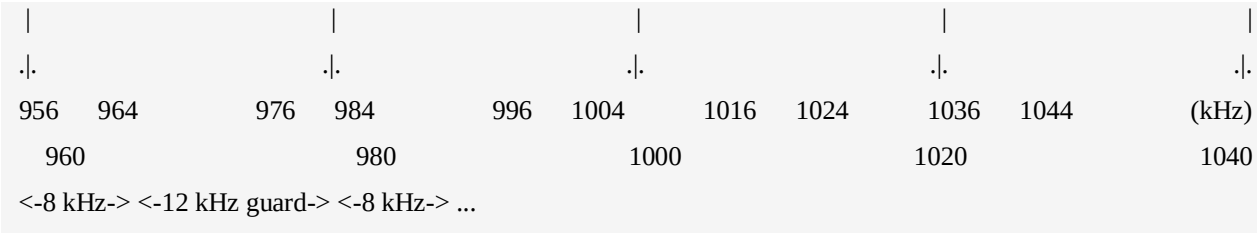
3.1 RF channel plan (five representative channels, AM, W = 4 kHz)

Channels are on a 20 kHz grid anchored at the 1 MHz candidate carrier. Each AM station occupies f_c ± 4 kHz.

Channel	Carrier f_c (kHz)	Lower sideband (kHz)	Upper sideband (kHz)	Occupied BW	Gap to next channel
1	960	956 – 960	960 – 964	8 kHz	12 kHz (964 to 976)
2	980	976 – 980	980 – 984	8 kHz	12 kHz (984 to 996)
3	1000	996 – 1000	1000 – 1004	8 kHz	12 kHz (1004 to 1016)
4	1020	1016 – 1020	1020 – 1024	8 kHz	12 kHz (1024 to 1036)
5	1040	1036 – 1040	1040 – 1044	8 kHz	–

Overlap check: the upper edge of channel n is f_n + 4 kHz and the lower edge of channel n+1 is f_n + 20 − 4 = f_n + 16 kHz. Since 16 kHz > 4 kHz, the spectra do not overlap and the guard band is 16 − 4 = 12 kHz in every case.

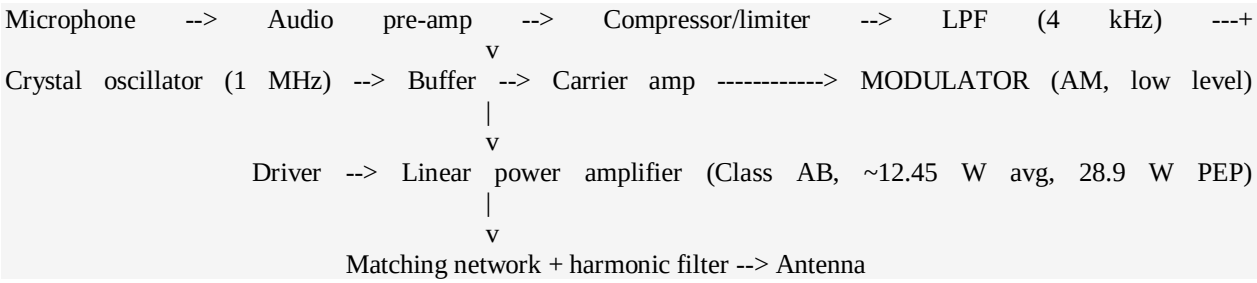
Sketch of the spectrum around the five channels (not to scale vertically; the tall line is the carrier and the small lines are the sidebands):



3.2 Channel spacing, guard bands and overlap in practice

- The ideal 8 kHz occupancy assumes the audio is strictly limited to 4 kHz. A real transmitter without an audio low pass filter can send components at 8–10 kHz, which produce sidebands up to 10 kHz away from the carrier. At 10 kHz the sidebands of neighbouring stations would just touch (10 + 10 = 20 kHz). So a steep 4 kHz audio low pass filter at the transmitter is part of the design.
- Over-modulation (μ greater than 1) creates harmonics and wide splatter that spills into neighbouring channels. The limiter keeps μ at 0.7 on peaks.
- Carrier frequency tolerance: a crystal oscillator with 10 ppm stability at 1 MHz drifts by 10 Hz, which is negligible compared with the 12 kHz guard band.
- Beat note: two carriers 20 kHz apart beat at 20 kHz, which is above the audio range, so there is no heterodyne whistle. With a 9 or 10 kHz spacing the beat would be audible, which is why the 20 kHz spacing is a safer choice for an educational demonstration.
- Receiver selectivity needs the guard band: the IF filter must give high attenuation at ±16 kHz from the carrier (where the first adjacent channel sideband begins), while still passing ±4 kHz.

3.3 Transmitter architecture



Function of each block:

- Pre-amp and limiter: raise microphone level and prevent the signal from exceeding $\mu = 0.7$ on peaks.
- 4 kHz low pass filter: limits the occupied bandwidth to 8 kHz.
- Crystal oscillator: gives the carrier frequency stability required by the channel plan.
- Modulator: forms $A_c[1 + \mu \cos \omega_m t] \cos \omega_c t$ at low power level.
- Linear power amplifier: raises power while keeping the envelope undistorted. A class C amplifier would clip the envelope, so a linear class AB/B stage is used, rated for 28.9 W PEP.
- Matching and harmonic filter: transfers power to the antenna and removes harmonics of 1 MHz that would fall in other channels.

Low-level modulation with a linear amplifier is chosen because it is simpler to build and adjust. High-level (plate or collector) modulation is more efficient, but it needs a modulating amplifier delivering roughly half the carrier power as audio, which is too heavy for a lab prototype.

3.4 Generation and detection blocks for AM, DSB-SC and SSB-SC

AM generation : m(t) + DC ---> [multiplier] <--- cos(wc t) ---> s(t) = Ac[1 + ka m(t)] cos(wc t)
AM detection : s(t) --> [diode] --> [RC low pass] --> [DC block] --> m(t)

DSB-SC gen. : m(t) --> [balanced modulator] <--- cos(wc t) ---> Ac m(t) cos(wc t)
DSB-SC det. : s(t) --> [product] <--- cos(wc t + phi) --> [LPF] --> (Ac/2) m(t) cos(phi)
(local carrier from Costas loop or squaring loop / pilot)

SSB-SC gen. : (filter method) m(t) --> [balanced mod] --> [sharp band pass filter] --> USB or LSB
(phase method) m(t) and its 90-degree shifted version m^(t) multiplied by cos and sin of the carrier, then subtracted (USB) or added (LSB)
SSB-SC det. : s(t) --> [product] <--- cos(wc t) --> [LPF] --> m(t)

3.5 VSB filtering concept and TV application

The response falls to half (−6 dB) at f_c and the two slopes are complementary, so the part of the lower sideband that is kept adds to the attenuated upper-side components to give a flat overall response. In television this shaping is done at the transmitter and partly at the receiver IF filter (Nyquist slope). With a 0.75 MHz vestige the video occupies about 5.25 MHz and the sound carrier is placed 4.5 MHz above the video carrier in NTSC, so the full channel is 6 MHz.

3.6 RF bandwidth and spectrum utilisation

Scheme	Bandwidth per station	Stations in 1060 kHz at 20 kHz spacing	Stations if spacing equals bandwidth (no guard)
AM / DSB-SC	8 kHz	53	132
VSB-SC (4.5 kHz)	4.5 kHz	53	235
SSB-SC	4 kHz	53	265

With the specified 20 kHz spacing the number of stations is fixed at 53 (52 on the 1 MHz grid) whatever the scheme, and the AM channel uses 8/20 = 40 % of its slot. If the spacing could be reduced, SSB would give the best utilisation, but a practical spacing is larger than the bandwidth because of the guard band needed by real filters.

Design Decision 2

I recommend the 52-channel plan with carriers at 560 + 20n kHz up to 1580 kHz (the five representative channels being 960, 980, 1000, 1020 and 1040 kHz), using a crystal-controlled low-level AM transmitter with a 4 kHz audio filter and a linear class AB power amplifier.

The trade-off is spectrum efficiency against complexity. AM uses only 40 % of each 20 kHz slot, and SSB could use 20 %, but SSB would need sharp filters or wideband phase networks in the transmitter and synchronous detectors in every receiver. Since the channel spacing is already fixed, the unused 12 kHz acts as a guard band that makes the filtering requirements simple and gives a margin for oscillator drift.